MEGATHON 2026 · PROBLEM STATEMENT

Accessible AI for All

A multilingual accessibility layer for AI assistants

IIIT Hyderabad · 10–11 October 2026 · For student teams



Concept illustration

AI assistants assume you can see, hear, speak, type and read fluent English

Millions in India do not fit that assumption: people who are deaf or hard of hearing, blind or low-vision, with speech or hand-motor impairments, and older adults, many working mainly in an Indian language.

Input gap

The tool itself is hard to use.

Output gap

Answers arrive as dense text, tables, charts or code that are hard to consume. This carries the most weight.

Pick any one of three use cases



1. Restaurant ordering

Start from a physical menu card. Understand items, descriptions and ingredients, order the food, give feedback and interact with staff.



2. Shopping Experience

Take a reference e Commerce platform. Enable browsing, product details and specs, reviews and placing an order.



3. City signage

Understand traffic signs, ads, shop boards and bus boards, then seek guidance and navigate.

Extend the chosen use case with your own imagination.

USE CASE 1 OF 3

Restaurant food ordering

Start from a physical menu card and go all the way to the bill.

Scenarios to explore

- Photograph the menu and hear each dish read aloud
- Ask: is this vegetarian, Jain or nut-free?
- Explain unfamiliar dishes and ingredients simply
- Compare dishes by spice level and price, then order
- Call the waiter, ask for the bill, split payment
- Give spoken feedback or a complaint to staff

Blind-first: read the menu in the user's language, describe layout and prices, confirm the order aloud.



Concept illustration

USE CASE 2 OF 3

eCommerce shopping experience

Pick any reference platform and make shopping work without sight.

Scenarios to explore

- Browse categories and filter results by voice
- Hear specs as a short, structured summary
- Turn hundreds of reviews into pros, cons, verdict
- Describe product photos and size charts in words
- Compare two products in plain language
- Place, track or return an order in your language

Blind-first: real headings, labelled buttons, spoken price and delivery details, a confirmation step before paying.



Concept illustration

USE CASE 3 OF 3

Navigating city signage

Understand the signs around you, ask for guidance and find your way.

Scenarios to explore

- Photograph a signboard and hear it in your language
- Bus boards: which bus goes to the station?
- Read traffic signs and pedestrian signals aloud
- Shop and ad boards: what is it, is it open?
- Get step-by-step guidance to a nearby landmark
- Find exits, platforms and emergency signs

Blind-first: short spoken directions, distance and direction cues, and a prompt to retake the photo when text is unclear.



Concept illustration

Pick three conditions; Visual Impairment is mandatory condition to be addressed.

Condition	Barrier	Example adaptations
Visual (mandatory)	Cannot see screens, images or charts	Screen-reader output, voice-first use, descriptions
Hearing	Cannot hear audio or voice replies	Captions, transcripts, visual alerts
Speech	Voice input not recognised	Phrase boards, word prediction, tolerant ASR
Hand-motor	Typing, mouse or touch is hard	Voice, switch or single-key control, fewer steps
Ageing	Small text, complexity, low comfort	Large text, slow voice, step-by-step help

Each condition needs a specific, demonstrable feature, not a generic accessibility mode.

System should be able to infer the condition and act accordingly

What your prototype must do

Bhashini speech-to-speech

Speak in one Indian language, hear the answer in the same or another: ASR, translation, AI, translation, TTS. Shown live.

Accessible outputs

Explain-simpler control, real headings and lists, tables and code described in words, captions, adjustable text and speech speed.

English plus two Indian languages

Input and output, in all three.

Graceful failure

Handle misrecognised speech and network drops by asking users to confirm or repeat.

ASR and or OCR › **Translation** › **AI** › **Translation** › **TTS**

Out of scope: training an LLM from scratch, medical, legal or financial advice, single-language solutions, skipping Bhashini, special hardware. Must run on a normal laptop or Android phone.

What to submit and demo

Deliverable	Detail
Working prototype	Hosted link or runnable build, plus source repo with a clear README
Write-up/ Presentation	Max 7 Slides: conditions, user needs, design, Bhashini use, testing, limitations
Demo video	Max 4 minutes, captioned and narrated; optional deck of up to 8 slides

Live demo: about 5 minutes. A demo per condition, a live Bhashini exchange (judges may supply the prompt) and a raw versus accessible output side-by-side.

How teams are judged

Criterion	Weight
Accessibility of outputs, not just the interface	25%
Depth and quality of the three chosen conditions	20%
Bhashini integration and Indian-language quality	20%
User understanding and real-world impact	15%
Technical quality, reliability, graceful failure	10%
Presentation, documentation, accessible demo	10%



Bonus, up to 5 points: validation with real users with disabilities, more than three conditions done well, offline or low-bandwidth support, extra Indian languages.